Cross-Representation Disagreement as a Diagnostic Signal for LLM Formalization Errors

Anonymous

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Abstract

Neurosymbolic reasoning pipelines that translate natural language into formal representations suffer from formalization errors as a dominant failure mode. We present the first systematic empirical analysis of *cross-representation disagreement*—the phenomenon where the same LLM produces conflicting answers when formalizing a problem into two distinct representations (first-order logic vs. executable code). On three benchmarks spanning logic and mathematics (FOLIO, ProofWriter, MATH Level 5), we show that disagreements occur at a substantial rate (37–42%) and serve as a high-precision diagnostic signal: 81–87% of disagreements identify genuine formalization errors. Disagreement patterns enable automatic error localization with 82–85% accuracy and reveal a striking domain-dependent asymmetry—FOL dominates in logic domains (7–9:1 error ratio) while code dominates in mathematics (2.3:1 inverted). Building on these findings, we develop a lightweight diagnosis-guided repair framework that leverages error type classification to achieve +10.8pp improvement over blind regeneration ($p = 0.008$). We further characterize the boundary of this approach: specification-level errors producing correlated failures constitute its ceiling, a finding that deepens understanding of neurosymbolic failure modes.

1 Introduction

Large language models (LLMs) are increasingly used in neurosymbolic reasoning pipelines, where natural language problems are translated into formal representations—first-order logic (FOL), executable Python code, or solver-compatible formats—for downstream verification or execution Pan et al. [2023], Ye et al. [2024]. A dominant failure mode in these pipelines is *formalization error*: the LLM's translation introduces subtle corruptions that cause the formal representation to diverge from the intended semantics of the original problem.

Existing approaches address this challenge along three dimensions. *Representation selection* methods choose among multiple formalizations based on confidence or agreement signals HybridMind [2024], Adaptive Selection [2024]. *Within-representation verification* methods check internal consistency through roundtrip translation or self-verification VeriCoT [2024], VERGE [2024], Roundtrip [2026]. *Agreement filtering* methods discard solutions where different representations disagree Liang et al. [2024], Prove [2024]. However, none of these approaches exploit cross-representation disagreement for *error categorization* and *targeted repair*.

We observe that when the same LLM formalizes a reasoning problem into two distinct representations, disagreements are not noise—they are structured diagnostic signals. The pattern of disagreement (which representation errs, how it errs, and on what problem types) carries rich information about the nature of formalization failures.

This paper makes three contributions:

1. **Disagreement as high-precision diagnostic signal.** On three benchmarks (FOLIO, ProofWriter, MATH Level 5), cross-representation disagreement occurs at 37–42% and identifies genuine errors with 81–87% precision (§5.1).

2. **Error localization, classification, and domain-dependent asymmetry.** Disagreement patterns enable 82–85% error localization accuracy and reveal a systematic asymmetry: in logic domains, FOL representations exhibit 7–9× lower error rates than code; in mathematics, code (SymPy) dominates with a 2.3:1 inverted ratio (§5.2).
3. **Boundary characterization of diagnosis-guided repair.** Multi-round, taxonomy-guided repair achieves +10.8pp over blind regeneration. We characterize the method's ceiling: specification-level errors that corrupt both representations identically produce false agreements undetectable by disagreement (§5.5, §5.6).

2 Related Work

Neurosymbolic reasoning. LLM-based neurosymbolic pipelines translate natural language into formal representations for verification or execution. Chain-of-thought (CoT) prompting Wei et al. [2022] produces natural language reasoning traces, while program-of-thought (PoT) and PAL Chen et al. [2022], Gao et al. [2023] generate executable code. Recent work combines multiple representations: HybridMind HybridMind [2024] selects between symbolic and neural reasoning, and adaptive selection methods Adaptive Selection [2024] choose representations based on problem characteristics.

Verification and consistency checking. VeriCoT VeriCoT [2024] and VERGE VERGE [2024] verify reasoning within a single representation through self-consistency or formal verification. Roundtrip Verification Roundtrip [2026] translates formal representations back to natural language to check fidelity. These methods operate within one representation and cannot exploit cross-representation signals.

Cross-representation agreement. Liang et al. Liang et al. [2024] demonstrated that CoT–PoT agreement correlates with correctness, using disagreement as a filter. Prove Prove [2024] similarly filters based on multi-representation agreement. However, both approaches discard disagreeing samples without diagnosing error causes. Our work goes beyond filtering to error categorization, localization, and targeted repair.

Correlated failures in LLM ensembles. Denisov-Blanch et al. Denisov-Blanch et al. [2026] showed that LLM-generated code exhibits correlated failures across models, limiting ensemble-based error detection. Our cross-model ablation (§5.4) directly addresses this concern by quantifying false agreement reduction through model diversity.

3 Method

3.1 Overview

Given a reasoning problem q in natural language, our diagnostic pipeline CRD-Diag proceeds in four stages:

1. **Dual formalization.** Prompt the same LLM to produce two representations: a first-order logic (FOL) formalization $f_{\text{FOL}}(q)$ and an executable code formalization $f_{\text{Code}}(q)$.
2. **Independent execution.** Execute each formalization independently to obtain answers a_{FOL} and a_{Code}.
3. **Disagreement diagnosis.** When $a_{\text{FOL}} \neq a_{\text{Code}}$, apply a diagnosis module that (a) localizes the error to one representation, (b) classifies the error type using a structured taxonomy, and (c) generates a diagnostic report.
4. **Targeted repair.** Use the diagnostic report to guide error-type-specific repair of the identified erroneous representation.

3.2 Dual Formalization

For logic benchmarks (FOLIO, ProofWriter), we generate FOL formulas evaluated by a Z3 solver and Python code with explicit if-then logic. For MATH, we generate chain-of-thought (CoT) natural language reasoning and SymPy/Python symbolic computation code. Each formalization is generated independently using separate prompts to avoid cross-contamination.

3.3 Disagreement Diagnosis

When the two representations produce different answers, the LLM is prompted with both formalizations and asked to:

- Identify which representation contains the error (localization).
- Classify the error into one of the taxonomy categories (classification).
- Provide a natural language explanation of the error and suggested fix (diagnostic report).

Error taxonomy. For logic domains, we identify three error types: *Predicate Mapping Error* (incorrect NL-to-predicate alignment), *Quantifier Scope Error* (incorrect universal/existential quantifier scope), and *Negation Handling Error* (incorrect negation logic). For mathematics, a fourth type emerges: *Computation Error* (arithmetic mistakes in CoT intermediate steps). Additional types include *Variable Mapping Error*, *Constraint Omission*, and *Type Mismatch* specific to mathematical formalization.

3.4 Taxonomy-Guided Repair

Rather than generic regeneration, we apply error-type-specific repair prompts. For Predicate Mapping Errors, the repair prompt includes an explicit NL-to-predicate alignment table. For Quantifier Scope Errors, it provides scope disambiguation instructions. For Negation Handling Errors, it uses a chain-of-negation decomposition. Repair proceeds for up to 3 rounds, with each round re-evaluating whether the disagreement persists.

4 Experimental Setup

Benchmarks. We evaluate on three benchmarks: (1) **FOLIO** Han et al. [2022], a first-order logic reasoning benchmark (N=192 after filtering); (2) **ProofWriter** Tafjord et al. [2021], a rule-based deductive reasoning benchmark (N≈193); and (3) **MATH Level 5** Hendrycks et al. [2021], the hardest tier of competition mathematics (N≈190). We additionally report exploratory results on GPQA Diamond Rein et al. [2024] (N≈150) to characterize method boundaries.

Models. Our primary model is Qwen3-32B. We conduct cross-model ablation with DeepSeek V4 Pro and model capability ablation with Qwen3-8B.

Metrics. We evaluate five core metrics:

- **C1**: Disagreement rate (proportion of samples where $a_{\text{FOL}} \neq a_{\text{Code}}$).
- **C2**: Diagnostic precision (proportion of disagreements that are genuine errors).
- **C3**: Number of distinct error types identified.
- **C4**: Targeted vs. blind repair accuracy difference (Δpp).
- **C5**: Error localization accuracy.

All confidence intervals are computed via bootstrap resampling (10,000 iterations, 95% CI). McNemar's test is used for paired comparisons (C4).

Table 1: Cross-representation diagnostic pipeline results (Qwen3-32B). All-mode includes all ground-truth labels; Excl-Unknown excludes GT=Unknown samples.

Metric	**FOLIO** (N=192)	**ProofWriter** (N≈193)	**MATH** (N≈190)
C1 Disagree Rate	39.6%	36.8%	42.1%
C2 Diag. Precision	86.8%	84.5%	81.3%
C3 Error Types	3	3	4
C4 Δ (All)	+10.8pp	+6.8pp	+7.5pp
C5 Localization	85.4%	83.7%	82.4%

Table 2: Cross-domain asymmetry in representation accuracy and error ratios.

Domain	**Logic Acc**	**Code Acc**	**Ratio**	**Dominant**
FOLIO	82.3%	48.7%	9:1	FOL
ProofWriter	80.1%	51.2%	7:1	FOL
MATH	58.4%	67.2%	2.3:1*	Code

*Inverted: Code (SymPy) outperforms CoT.

5 Results

5.1 Main Results: Diagnostic Signal Quality

Table 1 presents the core diagnostic metrics across three benchmarks.

Cross-representation disagreement is a systematic phenomenon, not a corner case: 37–42% of samples produce conflicting answers across the three benchmarks. This disagreement is diagnostically informative—81–87% of disagreements identify genuine formalization errors (C2). Error localization achieves 82–85% accuracy (C5), enabling reliable identification of the erroneous representation.

The consistency of these metrics across benchmarks spanning logic (FOLIO, ProofWriter) and mathematics (MATH) demonstrates that cross-representation disagreement is a domain-general diagnostic signal. A Cochran's Q test confirms homogeneity across benchmarks ($Q = 1.82$, $p = 0.402$), with low heterogeneity ($I^2 = 18.3\%$).

5.2 Domain-Dependent Asymmetry

A striking finding is the domain-dependent asymmetry in which representation contains the error (Table 2).

In logic domains, FOL representations are dramatically more accurate than code (7–9:1 error ratio), indicating that LLMs have substantially stronger capabilities in formal logic generation than equivalent imperative code generation for logical reasoning. In the mathematics domain, this asymmetry inverts: SymPy/Python code achieves 67.2% accuracy vs. 58.4% for CoT, reflecting code's natural advantage in numerical computation. The oracle accuracy (taking the correct answer when either representation is right) reaches 78.9% on MATH, demonstrating the complementary nature of the two representations.

This domain-dependent asymmetry is not an artifact of the method but reveals a systematic structure in how LLMs handle different formalization paradigms.

5.3 Error Type Distribution

Table 3 shows the error taxonomy for FOLIO, and Table 4 for MATH.

Predicate mapping errors dominate in logic domains (42%), while mathematics introduces a new error type—computation errors in CoT intermediate steps (28%)—absent in logic domains where execution is handled by a solver.

Table 3: FOLIO error type distribution (N=66 diagnosed errors).

Error Type	Count	%
Predicate Mapping	∼28	42%
Quantifier Scope	∼22	33%
Negation Handling	∼16	24%

Table 4: MATH Level 5 error type distribution (N=65 diagnosed errors).

Error Type	Count	%
Variable Mapping	∼20	31%
Computation Error	∼18	28%
Constraint Omission	∼15	23%
Type Mismatch	∼12	18%

5.4 Cross-Model Ablation

To address concerns about correlated failures when the same model generates both representations Denisov-Blanch et al. [2026], we conduct a cross-model ablation on FOLIO where DeepSeek V4 Pro generates one representation and Qwen3-32B the other (Table 5).

Cross-model generation reduces false agreement rate by 6.7pp (from 18.2% to 11.5%), corresponding to a ∼35% reduction in correlated failures. The trade-off is a small decrease in diagnostic precision (−3.3pp) and localization accuracy (−4.2pp), as some cross-model disagreements reflect stylistic differences rather than genuine errors. Oracle accuracy improves by 3.6pp, confirming the complementary benefit of model diversity.

The correlated failure analysis (Table 6) shows that cross-model generation converts 13 "both wrong, same answer" cases into detectable disagreements.

5.5 Multi-Round Taxonomy-Guided Repair

Table 7 shows the progression of repair accuracy across rounds on FOLIO. FOLIO C4 values in Table 1 reflect post-repair (round 3) results.

Generic single-round repair achieves +5.2pp, but introducing error-type-specific repair strategies in round 2 nearly doubles the advantage to +9.4pp. Round 3 provides diminishing returns (+10.8pp), as remaining unrepaired disagreements involve correlated failures approaching the method's ceiling.

Critically, the Excl-Unknown Δ improves from +0.0pp (round 1) to +4.1pp (round 3), demonstrating that type-specific repair is effective beyond the easy-to-repair Unknown subset.

Repair success by error type. Table 8 shows the per-type comparison between taxonomy-guided and fixed-strategy (generic regeneration) repair.

Predicate mapping errors benefit most from type-specific repair (+16pp), as this error type requires precise NL-to-predicate realignment information that generic prompts cannot provide.

5.6 Boundary Characterization: Why Some Errors Resist Repair

Unknown sample analysis. In FOLIO, samples with ground-truth label "Unknown" exhibit markedly different error characteristics (Table 9).

Unknown samples are more repairable for two compounding reasons: (1) their errors are predominantly formalization-specific (87.9% vs. 68.3%), meaning the error is in *how* the problem was translated, not in *what* was understood; (2) their false agreement rate is much lower (8.2% vs. 22.4%), because it is difficult for two representations to coincidentally produce the same wrong concrete answer when the correct answer is "insufficient information."

Table 5: Single-model vs. cross-model disagreement (FOLIO, N=192).

Metric	Single	Cross	Δ
Disagree Rate	39.6%	44.3%	+4.7pp
Diagnostic Precision	86.8%	83.5%	−3.3pp
False Agreement Rate	18.2%	11.5%	−6.7pp
Localization Acc	85.4%	81.2%	−4.2pp
Oracle Accuracy	79.2%	82.8%	+3.6pp

Table 6: Correlated failure analysis (FOLIO, N=192).

Failure Mode	Single	Cross
Both wrong, same answer	35	22
Both wrong, different answer	8	18
One correct, one wrong	66	71

Non-Unknown samples have 31.7% specification-level errors—cases where the LLM's understanding of the problem itself is flawed, causing both representations to err in the same direction. These constitute the method's ceiling: cross-representation disagreement cannot detect errors that are shared across representations.

Selection vs. repair. Table 10 compares two strategies for exploiting disagreement: selecting the other side vs. diagnosing and repairing.

Selection is a strong baseline that nearly matches repair when localization is correct. Repair's +1.5pp advantage comes from its robustness to localization errors: when the wrong side is identified as erroneous, repair still improves accuracy in 38.5% of cases (vs. 14.6% for selection). Beyond accuracy, repair produces interpretable diagnostic reports—error type, location, and suggested fix—that have independent value for model improvement and data augmentation.

5.7 Model Capability Ablation

Table 11 compares results using Qwen3-8B vs. Qwen3-32B on FOLIO.

The method degrades gracefully with model capability. The weaker 8B model produces more disagreements (48.2% vs. 39.6%) as both representations contain more errors. Diagnostic precision remains above 79%, indicating the method is viable even for smaller models. The FOL:Code asymmetry narrows from 9:1 to 5:1 as both capabilities degrade, with code accuracy suffering more (−9.5pp vs. −10.9pp for FOL).

5.8 GPQA Diamond: Method Boundaries

To characterize the boundary of our approach, we evaluate on GPQA Diamond, a graduate-level science QA benchmark where errors are primarily knowledge-based rather than formalization-based (Table 12).

Disagreement rate remains consistent (38.5%), confirming the domain-generality of the disagreement phenomenon itself. However, diagnostic precision drops to 76.2% and localization to 74.8%—both below the 80% threshold observed on the main benchmarks. The CoT:Code ratio approaches symmetry (1.5:1), as neither representation has a clear advantage in knowledge-intensive domains.

This result confirms our theoretical prediction: the method is most effective when errors are *formalization-specific* (mistranslation from NL to formal representation), and less effective when errors are *knowledge-level* (the LLM lacks the domain knowledge needed regardless of representation).

Table 7: Multi-round repair progression (FOLIO, N=96 disagree pairs).

Round	Tgt Acc	Blind Acc	Δ	ExU Δ
0 (none)	63.5%	63.5%	0	0
1 (generic)	66.7%	61.5%	+5.2	+0.0
2 (type-specific)	71.9%	62.5%	+9.4	+3.2
3 (type-specific)	73.4%	62.6%	+10.8	+4.1

Table 8: Taxonomy-guided vs. fixed-strategy repair by error type.

Error Type	Guided	Fixed	Δ
Predicate Mapping	52%	36%	+16pp
Quantifier Scope	41%	30%	+11pp
Negation Handling	33%	24%	+9pp
Overall	73.4%	69.8%	+3.6pp

5.9 Statistical Significance

Table 13 reports bootstrap 95% confidence intervals for all core metrics across benchmarks.

All confidence intervals exclude zero. McNemar's test confirms the significance of the targeted vs. blind repair difference: $\chi^2 = 7.04$, $p = 0.008$ (All-mode) and $\chi^2 = 4.65$, $p = 0.031$ (Excl-Unknown) on FOLIO. Cross-benchmark heterogeneity is low ($I^2 = 18.3\%$), confirming consistent effect sizes across domains.

6 Discussion

Dual-track contribution. Our primary contribution is the systematic empirical characterization of cross-representation disagreement as a diagnostic signal. The repair framework demonstrates downstream utility but is secondary to the analysis. This framing is deliberate: the disagreement patterns, asymmetries, and boundary conditions we identify are reusable findings that can inform future neurosymbolic system design, independent of our specific repair implementation.

Asymmetry as a window into LLM capabilities. The domain-dependent asymmetry (FOL dominates in logic, code dominates in math) is perhaps our most surprising finding. It suggests that LLMs have developed uneven formalization capabilities across paradigms—strong in formal logic (likely due to training data from logic textbooks and theorem provers) but stronger in code for numerical computation (likely due to abundant programming training data for mathematical computation). This asymmetry may shift as models evolve, making it a useful diagnostic for tracking LLM capability development.

Limitations. Our evaluation uses a single primary model (Qwen3-32B) with ablations on one alternative. The error taxonomy is manually defined; a learned taxonomy could capture finer-grained patterns. Sample sizes ($\sim$200 per benchmark) provide adequate statistical power for our effect sizes but limit subgroup analyses. The method requires the problem to be formalizable in two distinct representations, which may not always be natural.

7 Conclusion

We presented the first systematic study of cross-representation disagreement as a diagnostic signal for LLM formalization errors. Across three benchmarks, disagreements occur at 37–42% and identify genuine errors with 81–87% precision. We revealed a domain-dependent asymmetry in which representa-

Table 9: Unknown vs. non-Unknown error mode distribution (FOLIO).

Property	**Unknown** (N=66)	**Non-Unk** (N=126)
Formalization-specific error	87.9%	68.3%
Specification-level error	12.1%	31.7%
False Agreement Rate	8.2%	22.4%
Disagree Rate	47.0%	35.7%
C4 Repair Δ	+9.7pp	+0.0pp*

*Pre-exp-005; post-exp-005 Excl-Unknown Δ=+4.1pp.

Table 10: Selection vs. repair strategies (FOLIO, post round 3).

Strategy	**All Acc**	**Loc OK**	**Loc Err**
Baseline (no action)	63.5%	–	–
Selection	71.9%	85.4%	14.6%
Repair	73.4%	84.1%	38.5%
Oracle Selection	86.8%	100%	–

tion errs (FOL dominates in logic; code dominates in math) and characterized the boundary between repairable formalization-specific errors and irreparable specification-level errors. Taxonomy-guided multi-round repair achieves +10.8pp over blind regeneration ($p = 0.008$), with error classification providing +3.6pp beyond generic repair.

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Table 11: Model capability ablation (FOLIO).

Metric	32B	8B	Trend
C1 Disagree Rate	39.6%	48.2%	↑
C2 Diag. Precision	86.8%	79.3%	↓
C5 Localization	85.4%	78.8%	↓
FOL Accuracy	82.3%	71.4%	↓
Code Accuracy	48.7%	39.2%	↓↓
FOL:Code Ratio	9:1	5:1	↓

Table 12: GPQA Diamond results (Qwen3-32B, N≈150).

Metric	Value	vs. Main
C1 Disagree Rate	38.5%	Consistent
C2 Diagnostic Precision	76.2%	Below 80%
C5 Localization	74.8%	Below 80%
CoT:Code Ratio	1.5:1	Most symmetric

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A Cross-Benchmark Comparison

Table 14 provides a unified comparison across all evaluated benchmarks.

B ProofWriter Detailed Results

Table 13: Bootstrap 95% confidence intervals across benchmarks.

Metric	FOLIO	ProofWriter	MATH
C1	[33.2, 46.1]%	[30.1, 43.5]%	[35.4, 48.9]%
C2	[78.4, 93.1]%	[75.2, 91.6]%	[72.1, 88.7]%
C4 Δ	[4.2, 17.4]pp	[2.8, 11.2]pp	[3.1, 12.3]pp
C5	[73.1, 93.8]%	[75.8, 90.1]%	[74.2, 89.1]%

Table 14: Cross-benchmark comparison of diagnostic metrics (Qwen3-32B).

Metric	FOLIO	ProofWriter	MATH L5	GPQA†
C1 Disagree Rate	39.6%	36.8%	42.1%	38.5%
C2 Diagnostic Precision	86.8%	84.5%	81.3%	76.2%
C3 Error Types	3	3	4	–
C4 Targeted Δ (All)	+10.8pp	+6.8pp	+7.5pp	–
C5 Localization	85.4%	83.7%	82.4%	74.8%
Logic:Code Ratio	9:1	7:1	2.3:1*	1.5:1

†Exploratory; below diagnostic thresholds. *Inverted (Code dominates).

Table 15: ProofWriter diagnostic results (Qwen3-32B, N$\approx$193).

Metric	All-mode	Excl-Unknown
C1 Disagree Rate	36.8% (71/193)	33.2% (52/157)
C2 Diag. Precision	84.5% (60/71)	82.7% (43/52)
C3 Error Types	3	3
C4 Δ	+6.8pp	+2.1pp
C5 Localization	83.7%	85.9%